STAT 330-001

40142034 65

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Test: 641566 Form: A Wadsworth, Spencer Gordon

Entry: 07:43pm 10/21/2026 Checked out:

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SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034

065

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

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Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Lasso and ridge regression are referred to as shrinkage methods because they can shrink or reduce certain coefficients of predictors so they have less effect. This is accomplished by adding bias to shrink specific coefficients.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

4/100 = .04

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right). = .48 = \frac{2.4^2}{3+1}\left(\frac{.25}{.75}\right)$$

.48 > .04

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

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(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{B} = (X^T X)^{-1} X^T y$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is the mean coefficints while β is the estimated coefficints.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

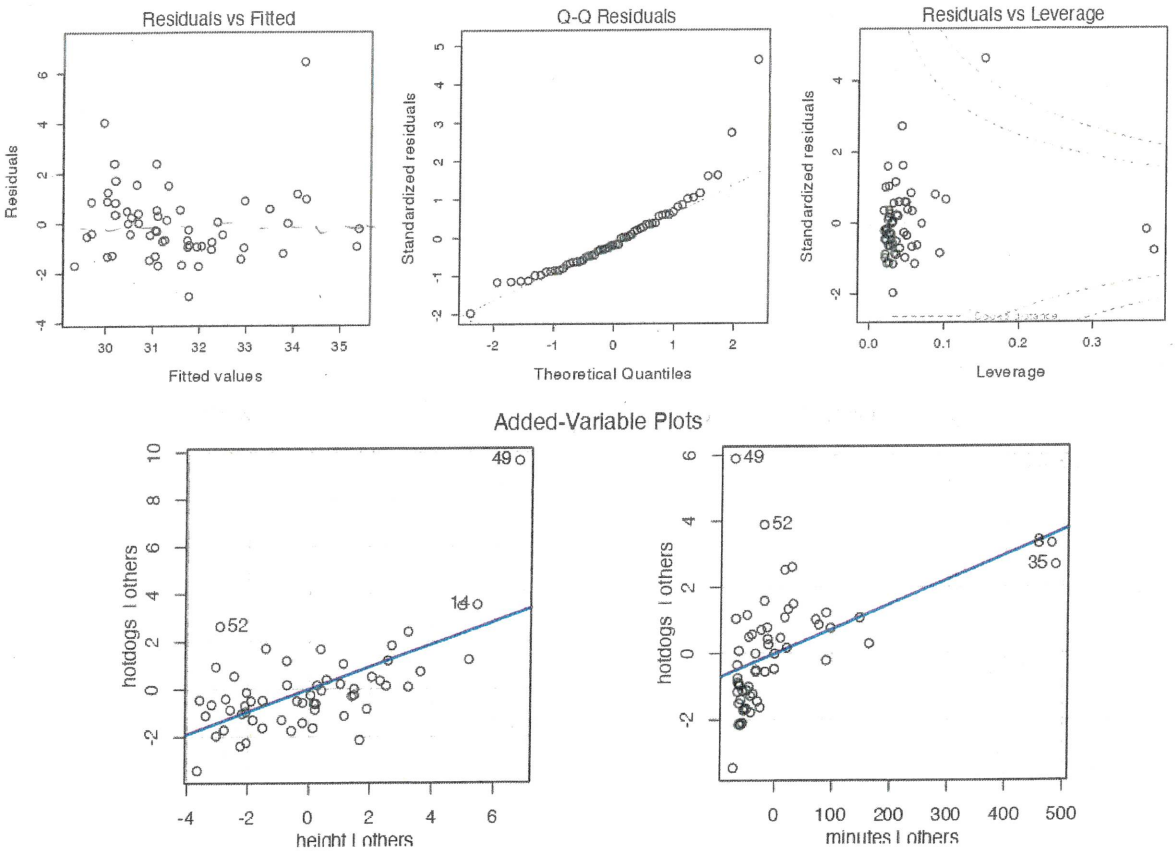
Least squares is used to estimate regression coefficints by finding the difference between the observed values - expected values and squaring them to find the errors in the data compared to our estimated values.

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$



Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

β_1(height) = .472892 which means holding all other predictors constant for every 1 increase in inches of height the predicted response, hot dogs eaten in 10 min, increases by .472892 hotdogs on average.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

$.007417 \pm .001871 \cdot 3.963$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Because the residuals vs fitted graph shows data clustered around a horizontal y line the relationship between the explanatory and response variables appear linear.

(B) There is independence between each observation

We would need more data on how each observation was conducted

(C) The errors are normally distributed

Because the Shapiro-Wilk p-value ($2.4Be-8$) $< .05$ there is evidence against the data being normally distributed.

(D) The errors have constant variance (homoscedasticity)

Because the Breusch-Pagan p-value ($.071$) $> .05$ there is not sufficient evidence against constant variance

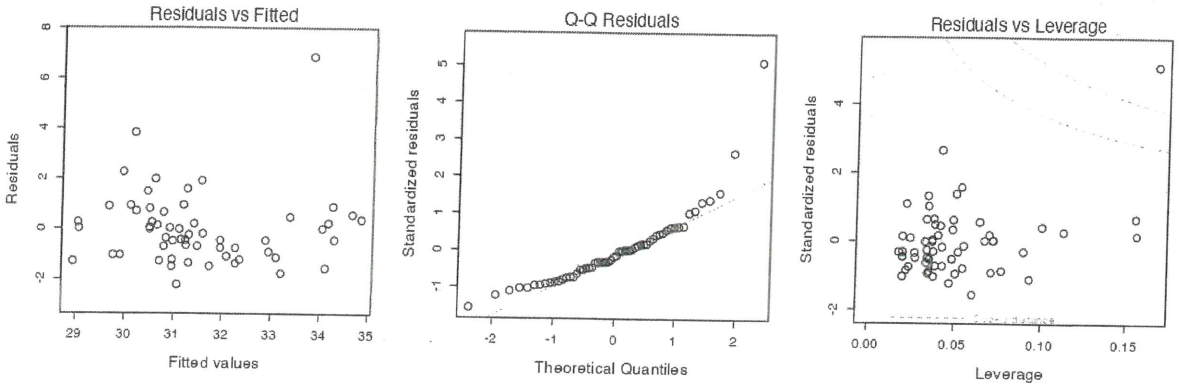
(E) There are no unduly influential points

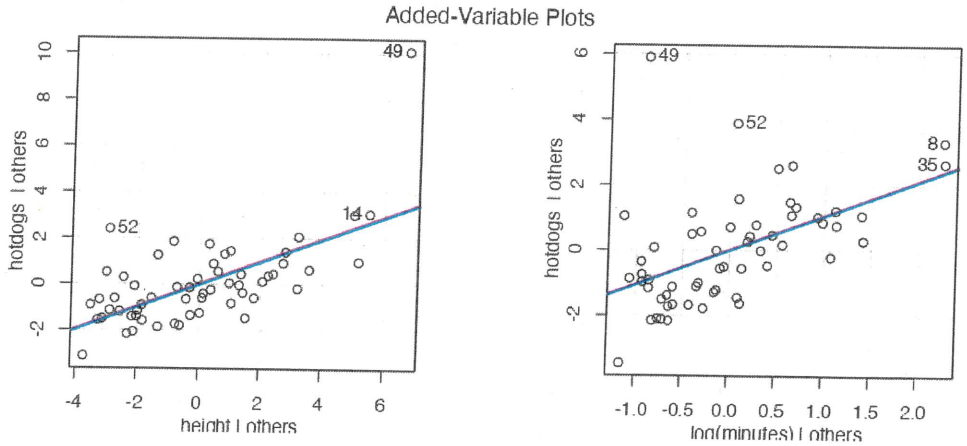
$4/58 = .06896657$ (cook's distance screener)

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

Even though model two has evidence against equal variance (Model 2 Breusch-Pagan test's p value < .05), that alone is not sufficient evidence to deem model 1 is superior as there are other tests and factors involved in deciding if one model is a superior model than another.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Disadvantage: Model 2 has evidence against equal variance.

Advantage: Model 2 has a lower CV RMSE which means it is a better model for predicting.

Question 16: State the estimated value of β_2 and provide an interpretation in context

β_2 (minutes) = 1.06702 log(x) so $\beta/100 = .0106702$

Holding height constant with every 1 minute increase in weekly training the predicted response, hot dogs eaten in 10 minutes, increases by .0106702 hot dogs.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

B_0 B_1 height B_2 minutes

$-6.05689 + .41834(68) + 1.06702 \log(79) = 31.813$ hotdogs in 10 minutes

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

$df = n - p - 1 = 58 - 2 - 1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.813 \pm 2.004 \cdot 1.485 = (28.83710, 34.78898)$

Part III - Application

Questions 20-24 concern the following dataset:

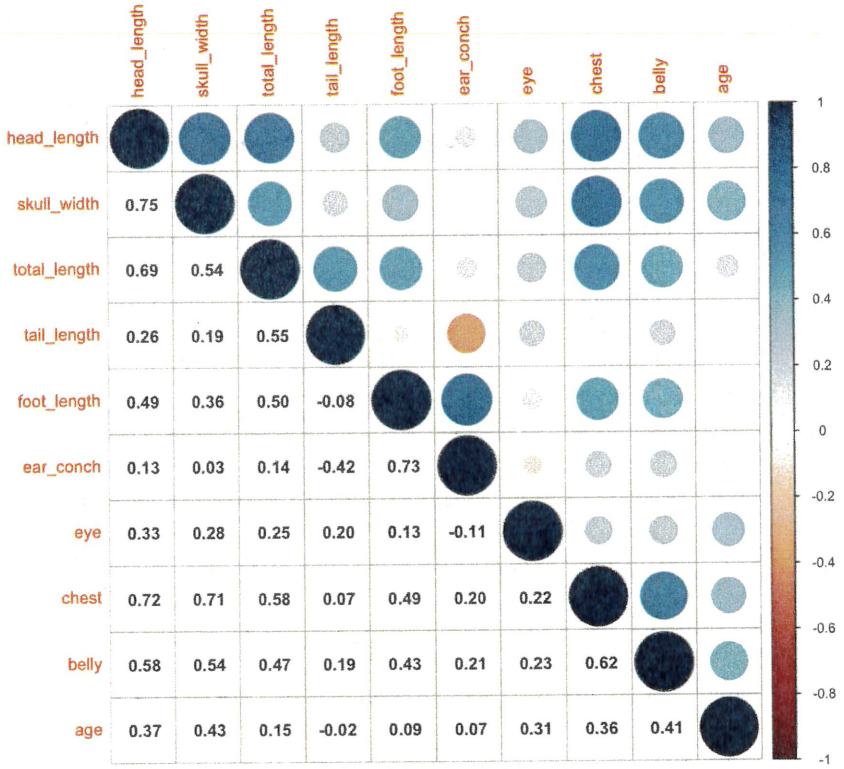
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

.16559 ± .10599(2.01)

.16559 ± .10599(2.0)

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% Confident that the true mean of β_{belly}, the coefficient on belly width, lies between (-.023, .443). Because 0 is in the interval we cannot conclude belly width is significant in determining age.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The f test tests if at least one predictor has a non zero coefficient, in this case the f test concludes there is at least one, but it doesn't test each predictor individually. That is to say that even though each predictor when tested individually are found to not have a significant effect, together at least one is significant.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

This could be because of multicolinearity

Question 22: The best subsets algorith compares all combinations of 2 predictors. It determines which pair of predictors to pick by using AIC criteria which balances fit and complexity when deciding which model to select.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

$$1 - (1-R^2)\frac{n-1}{n-p-1}$$

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

100 > 80 20

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

CV steps: split / \ fit test

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

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Test: 641630 Form: A Wadsworth, Spencer Gordon

Entry: 08:01pm 10/21/2026 Checked out:

Employee: Elijah B.

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(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

they shrink how much the X_j values affects predictions by biasing or 'shrinking' the $\hat{\beta}$ values towards zero. In LASSO, they can also shrink the $\hat{\beta}$'s to zero, which makes a simpler model.

$\frac{4}{n} = \frac{4}{100} = 0.04$

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use Cooks Distance

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{2.4^2}{3+1} = \frac{5.76}{4} = 1.44$

$\frac{0.25}{1-0.25} = 0.33$

$1.44(0.33) = 0.48$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

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Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\widehat{\beta} = (X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is the true effects of the predictor variables on the response. the true β's exist in the world. $\widehat{\beta}$ is our estimate of the true effects of predictor variables based on the data we have gathered. Our $\widehat{\beta}$'s are attempting to estimate the true β's.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

We choose the $\widehat{\beta}$ values that decrease the squared residual values. This helps us estimate regression coefficients Bc we choose the model that fits our data the best (by decreasing every value's distance from our predicted $\hat{y}$ as much as possible). this works bc the true β values ended up with the actual results, so we use our sample to try and estimate what β values resulted in our observed sample.

$SSE(\beta) = (Y - X\beta)^T(Y - X\beta)$

$= Y^TY - 2\beta^TX^TY + \beta^TX^TX\beta$

$\frac{\partial SSE(\beta)}{\partial \beta} = -2X^TY + 2X^TX\beta = 0$

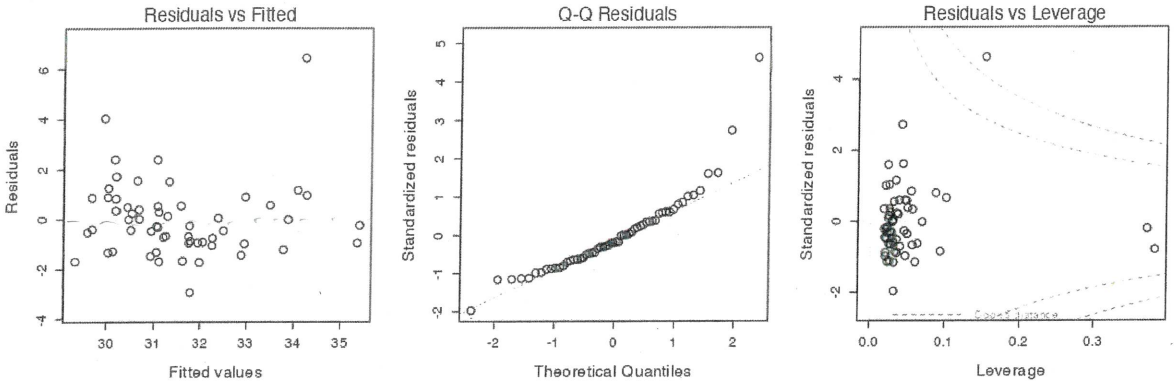
$X^TX\widehat{\beta} = X^TY \rightarrow \widehat{\beta} = (X^TX)^{-1}X^TY$

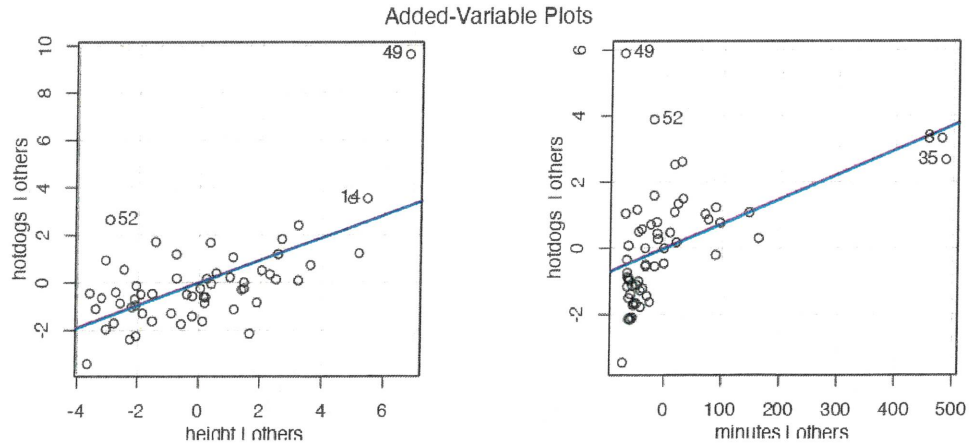
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimated value of β_1, $\hat{\beta}_1 = 0.472892$. This means that holding the amount of minutes contestents spend training per week constant, a 1 inch increase in height is associated with an estimated 0.472892 increase in the amount of hot dogs eaten by the contestent in 10 minutes.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

$0.007417 \pm (3.963)(0.001871)$

$(0, 0.01)$

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

no bc $\hat{\beta}_1 = 0.4729$ which is not the same as $\beta_1 = 0.4729$

0.01

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Besides some funneling (which affects equal variance) there does not appear to be discernable curvature/trends in the residuals vs. Fitted plot that would indicate that linearity has been violated. In the hotdogs AV, the points roughly follow the linear line with only a single point far from the others in the right hand corner, the AV plot for minutes does give concern for whether there is a linear relationship between minutes & Hotdogs. A transformation for Minutes may be warranted.

(B) There is independence between each observation

We are not given much information about how the 58 ametur competitive eaters were chosen - whether they were randomly selected or not. There is reason for concern about whether this assumption was met since if family competed together they might not be independent or if contestants dont go all at the same time one could see the score of someone else and be motivated to eat more hot dogs than they would've.

(C) The errors are normally distributed

The Q-Q plot has points curving off the line on both sides, but especially the right. This is concerning because it indicates a possible extreme skew in the residuals. Combine this with a very small, significant Shapiro Wilk test and there is concerning evidence that the normality assumption is not met. With the probable skew, this could mean that the CI's & PI's will have inaccurate coverage, along with other concerns.

H_0: Residuals come from a normal Dist
H_A: Residuals Don't come from a normal Dist
Since the p-value (2.43e-5) is much smaller than α (0.05) we reject H_0. There is convincing evidence that the errors/residuals do not come from a normal Dist.

(D) The errors have constant variance (homoscedasticity)

There appears to possibly be some slight Funnelling in the residuals vs. fitted plot, but it is not enough to give solid evidence that the constant variance assumption has been violated, especially with a non-significant Brach-Pagan p-value. The funneling is worth noting, but shouldn't be enough to make CI's, PI's, or t- & F tests unreliable.

Breuch-Pagan test
H_0: Error variance is constant
H_A: Error variance depends on the predictors
bc our p-value is 0.071 which is $> \alpha (0.05)$ we fail to reject H_0. there is not convincing evidence that the error variance depends on the predictors

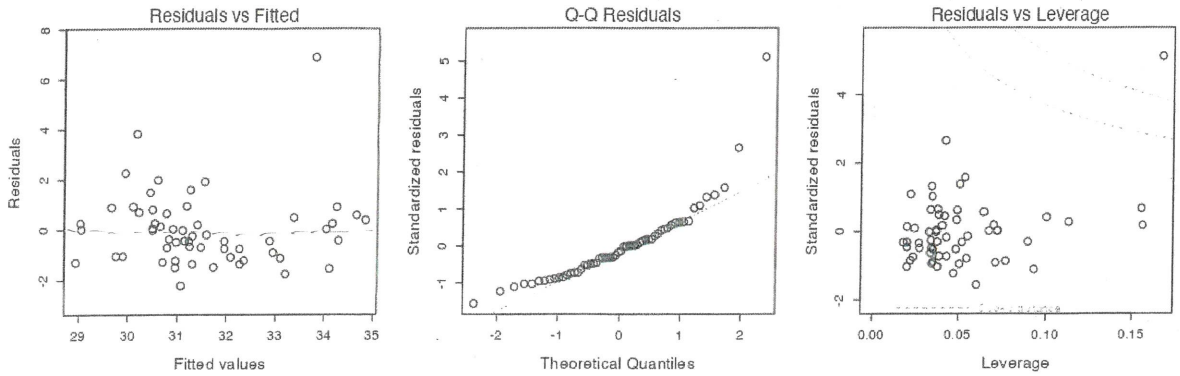
(E) There are no unduly influential points

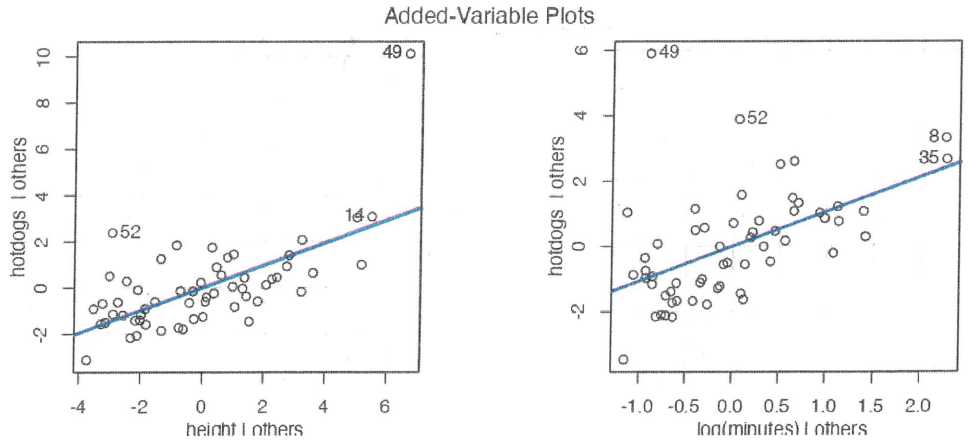
There is a point in the top right that crosses the Cook's Distance lines. This is a warning sign that we should investigate the observation and it's data quality since it appears to be unduly influential.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say that a model's superiority cannot be proved with one singular p-value. Although Model 2's fulfilment of the equal variance assumption may be 'worse', the transformation can help improve the model's ability to meet the other assumptions or could strengthen the model's predictive power which would probably make model 2 the stronger overal model.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages

- the AV plot for minutes is much improved, now giving evidence of a linear relationship between minutes & hotdogs
- The new CV RMSE value is smaller, giving evidence that model 2 predictive power is greater than for model 1.

- Normality is still a problem and the shapiro-wilk test's p-value is even lower than before
- Based on the funneling in the residuals vs. Fitted plot and the Breusch-pagan test's p-value, equal variance appears to be slightly more of a concern.

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\hat{\beta}_2 = 1.06702$ $\quad \frac{1.06702}{100} = 0.0106702$

Holding height constant, a 1% increase in minutes per week spent training for a contest is associated with an estimated increase of approximately 0.0106702 in mean hot dogs eaten in 10 minutes.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{y} = (-6.05589) + (0.48834)(68) + (1.06702)(\log(79)) + \varepsilon$

$\hat{y} = \boxed{31.81} + \varepsilon$

We would predict that he would be able to consume about 31.81 hot dogs in 10 minutes.

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$58 - 2 - 1 = 55$

$df = n - p - 1$

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

BC it's a prediction interval, we use this one

$est \pm t^* \cdot SE$

$31.81 \pm 2.004 \cdot 1.485$

$\boxed{(28.83, 34.79)}$ — 95% PI for the # of hotdogs my friend can consume in 10 minutes

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

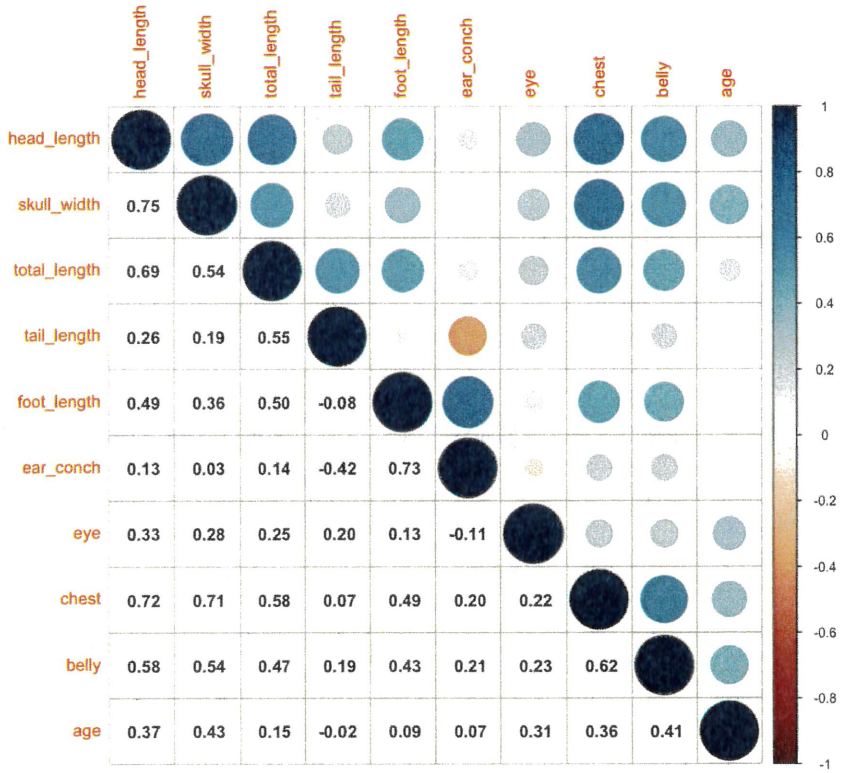
Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

Response

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

est ± t* · SE

0.16559 ± (2.01 · 0.10599)

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true belly width coefficient (or the true effect of belly width on our response variable age) is captured by the interval (-0.023, 0.443)

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

the model itself looks at all the variables combined, and since we know there is some level of multicollinearity, while one predictor may not individually account for the variation in the response, together the variables may be significant.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

It creates every possible model from every possible predictor combination and then uses the AIC penalty of $n\log(\frac{SSE}{n}) + 2k$ and chooses the model that minimizes the AIC penalty.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

____ Average RMSE across all test-training splits and report

____ Average the predictions and record their value

C Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

____ Initialize vectors for each model that you will use to store their RMSEs

____ Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 61

Otero Bazan, Lucero Maribell

250793085 blucerom 385-401-9673

Test: 641458 Form: A Wadsworth, Spencer Gordon

Entry: 07:14pm 10/21/2026 Checked out:

Employee: Ryan A.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034
041

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They are shrinkage methods because the lower total ____ error. They use λ & push slopes towards 0, the larger λ the more stronger the shrink or penalization

LASSO even pushes the coefficients down to zero which in a way gets rid of them & serves as a selection method & reduces variance & improves prediction.

Ridge additionally penalizes^2 whenever there are more predictors than observations. And improves out of sample predictions.

Both are

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right). = \frac{(2.4)^2}{3+1}\left(\frac{.25}{1-.25}\right) = \frac{5.76}{4}(0.33) = 0.4752$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

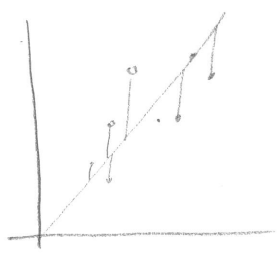
(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is the fitted coefficient

β is the true or parameter coefficient, often unknown

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

Least squares:



Least squares helps finding the slope at its best position in order to have the smallest possible sum of distance from the predicted point (in the slope) to the true observation

Having the smallest sum assures us that the distance or regiduals are setting the coefficients in the best possible way for the linear regression.

We get the residual squared & the sumed. We squared so that negatives & positives

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

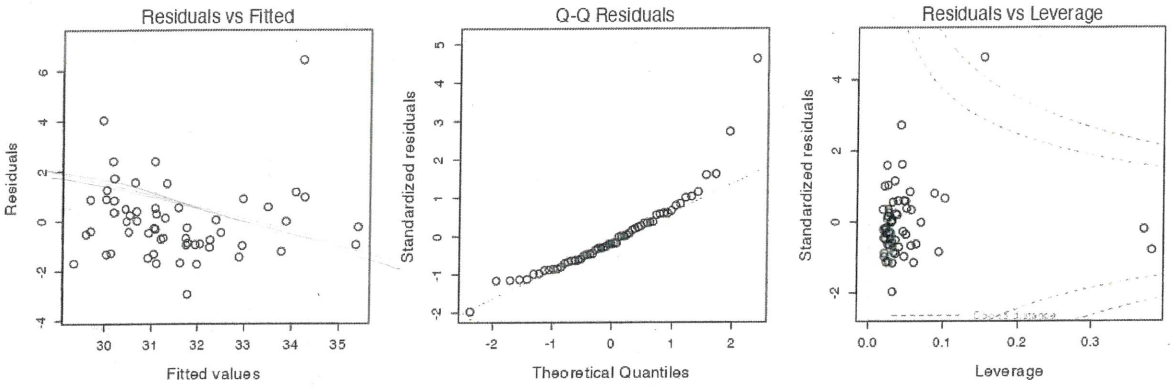
n=58

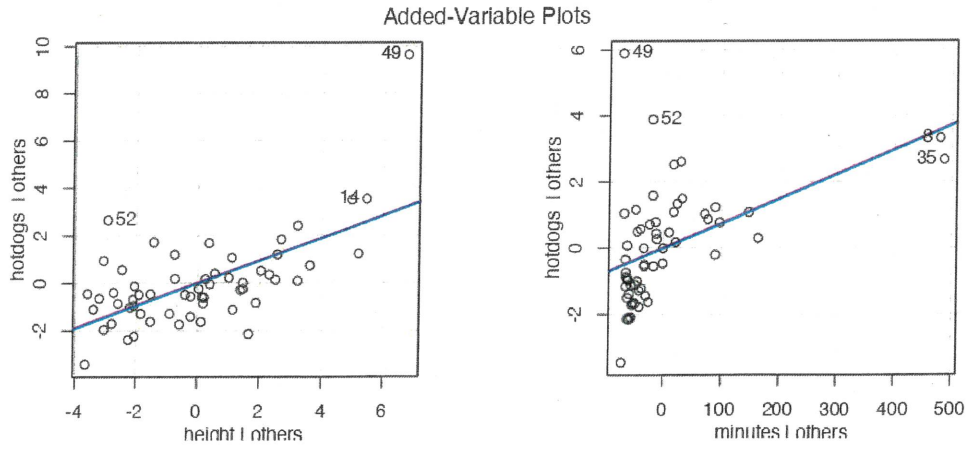
independency compromised?

4 largest counties

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\hat{\beta}_1 = 0.47$

Holding minutes (of training for a contest per week) Constant, a One inch increase of height has an average increase of 0.47 Hot dogs consumed in ten minutes

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$ β_0 is the intercept

(D) $\beta_1 = 0.4729$ should be $\hat{\beta}_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Based on the Residual vs Fitted plot, with which we can asses the linearity assumption, the data looks kind of linear, with some outliers & a bit of a funnel shape decreasing toward the right I would still consider linearity. The added variable plots indicate that height is more linear. Minutes have a funnel or pattern. So the assumption is violated

(B) There is independence between each observation

The data was gathered from people at a State fair & they came from the 4 largest counties, so some connection exists, but it might not be relevant to our variables. We don't know the sampling method but we can assume it was random.

(C) The errors are normally distributed

The Q-Q Residual plot shows a right skewed distribution & the Shapiro-Wilk p-value is significant. I would say I don't pass it because the plot shows some serious skewness & p value supports

(D) The errors have constant variance (homoscedasticity)

From residuals vs fitted I'd say I pass it the funnel shape is not very serous & the Breusch-Pagan p-value is 0.071 so is not significant

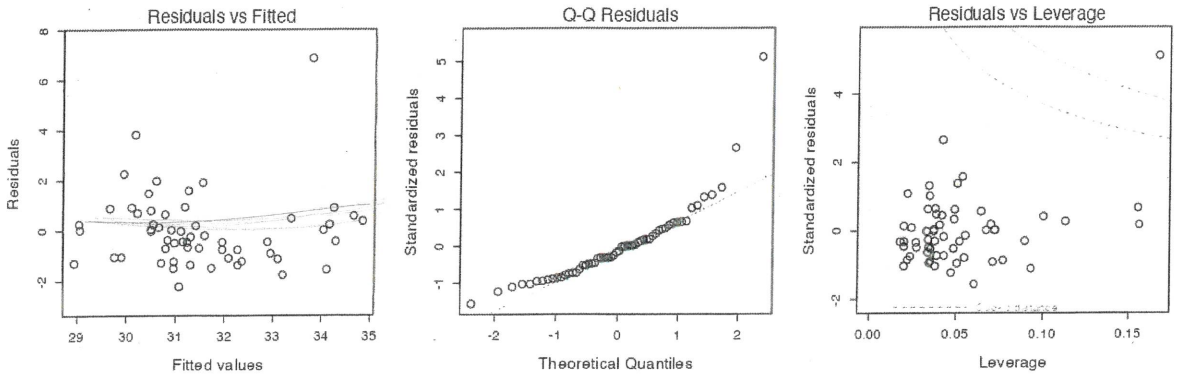
(E) There are no unduly influential points

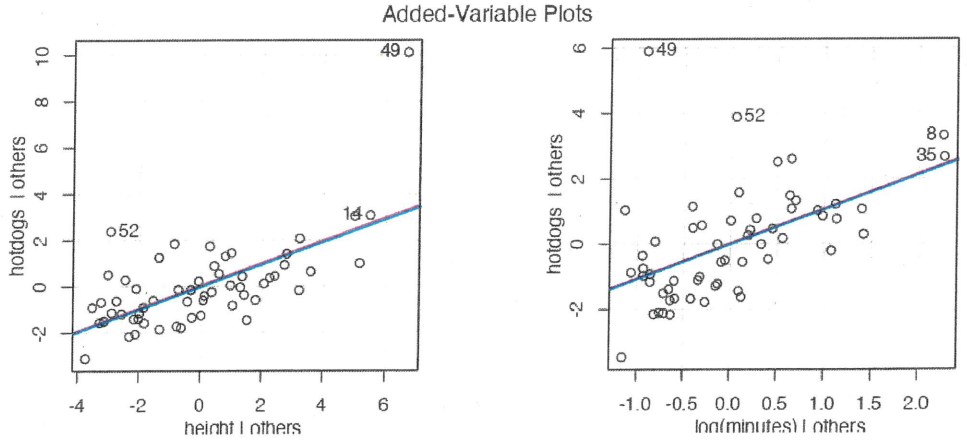
Cooks distance. From the residual vs. leverage, an observations are high in Standardize regidual (SR) related to the rest, and 2 observations a high in leverage relatively. They are however not any observation high on both, perhaps the point 5 high SR farther from all so that would be influential.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

The tests are good supporting, but in diagnostics the most important items are the visuals. Often a larger sample increases p-value from unimportant things. Thus, we need to trust our eyes and see that the plot does not really (Residuals vs. Fitted) show a significant difference in error variance, and also check all other diagnostics!

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages:
Better linearity as observed from the Added-Variable Plots of predictors & response considering other predictors. Minutes has a larger coefficient
CVRMSE decreased but is now CV error, Shapiro-Wilk p-value decrease
Reduced influence from observations identified in with Cook's distance

Disadvantages
Normality & Overall linearity stay the same
Breusch-Pagan Pvalue - decreased

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\hat{\beta}_2 = 1.067$

Holding height (in inches) constant, a 1% increase in minutes (training for a contest) corresponds 2.1386 units in mean of hot dogs consumed in 10 minutes

$$\frac{\beta_2}{100} = \frac{1.067}{100} = 0.01067$$

$+C\%$ in $x \rightarrow \beta_2 \log\left(1 + \frac{C}{100}\right) =$ ← I choose this lol

$$\beta_2 \log\left(1 + \frac{1}{100}\right) = 2.1386$$

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{\beta}_0 = -6.05589$

$\hat{\beta}_1 = 0.48834$

$\hat{\beta}_2 = 1.06702$

$\widehat{RMSE} = 1.66702$

$\text{Hotdogs}_i = \beta_0 + \beta_1 \text{height}_i + \beta_2 \log(\text{minutes}_i) + \epsilon_i$

$\widehat{\text{Hotdogs}}_1 = -6.05589 + (0.48834)(68) + (1.06702)(4.369) + 1.66702$

$33.20712 \quad 4.66181038$

$i = \text{friend} = 1$

$\text{height}_i = 68 \text{ in}$

$\text{minutes}_i = 79 \text{ min/week}$

$\log(79) = 4.369$

≈ 33.48

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

$n - p - 1 = 58 - 2 - 1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

PI: $\hat{y} \pm t^* \, SE$

$33.48 \pm 2.004 \cdot 1.485$

33.48 ± 2.97594

PI: $(30.50406, 36.45594)$

There is a 95% chance that the consumed hotdogs by my friend lie between 30.50 & 36.46

Wider than CI because of the 1 in the root & because we are predicting someone's response.

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

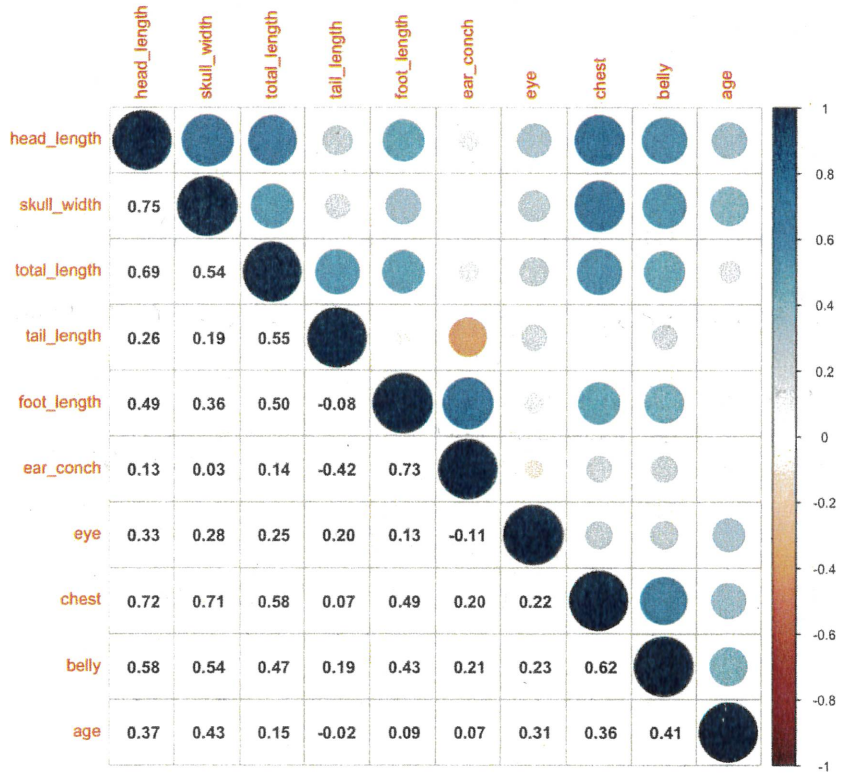
Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.

Not bad



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$\hat{\beta}_i \pm t^* \, SE$

$.16559 \pm (1.562)*(.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

If repeated sampling there is a 95% probability that the true coefficient of belly width would fall between -0.023 and 0.443

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

When measured individually the coefficients might not show their influence, perhaps because of multicollinearity, they are sharing their influence & that kind of hides them, but the F-test showing significance uses all variables to determine the effects.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best Subsets tests each possible combination of predictors or possible model (2^n) where n is the number of predictors. It is exhaustive algorithm of variable selection. AIC as a metric uses $= n \log(SSE/n) + 2K$, which penalizes the less important predictors (not as much as BIC) good if you consider having more predictors (or certain ones) unproblematic or the value is less than other metrics, and trade off is worth it

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

probably not, because adding predictors keeps adjusted R^2 as it is or lowers it. I am assuming this model has less predictors than originally so
So the original model had n predictors with a smaller (AR^2) adjusted R-squared, we took some out so AR^2 is less small or bigger, but not the highest since the model could continue taking out predictors, so is not the highest.

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

A Compute each model's AIC and BIC

D Fit each model using the testing data

A Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

D Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 60

Aragon, Trevor

357735835 taragon7 951-226-4244

Test: 641426 Form: A
Entry: 07:08pm 10/21/2026
Employee: Ashlyn T.

Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034
060

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They are referred to as "shrinkage" methods because the shrink coefficients to help narrow down the true predictors of the model

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(1.44) (.33) = .48

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model. ✗

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50. ✗

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model. ✗

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50. ✗

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (p+1) \times 1$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is the estimated value of a coefficient

while β represents the true value of the parameter

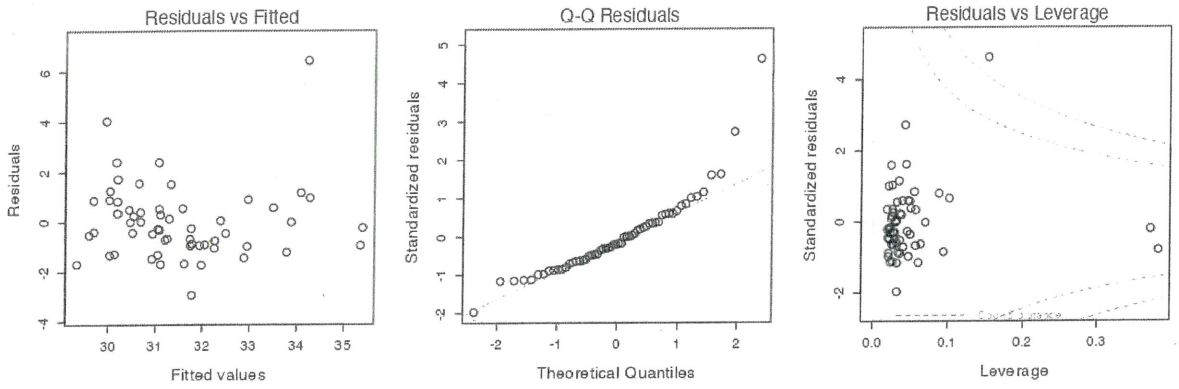
(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

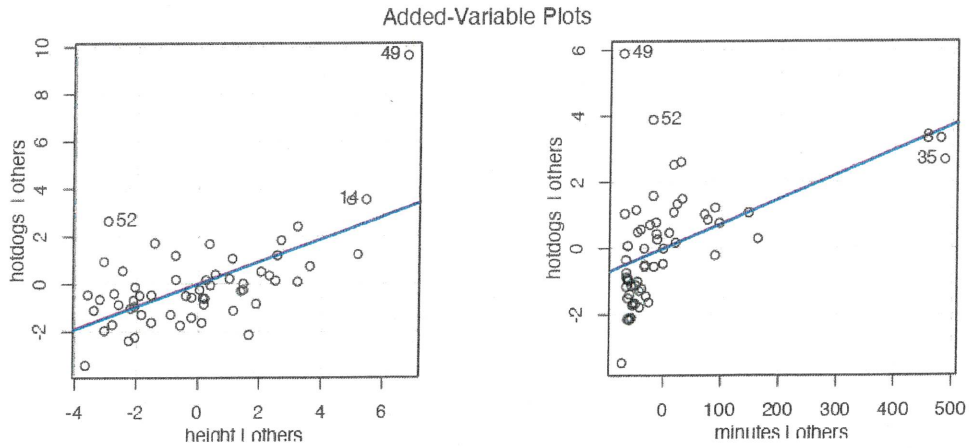
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\beta_1 = .472892$ Holding minutes constant a 1unit increase in height is associated with a 47.29% decrease in the conditional median of y

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0 .3963

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Based on Residuals vs Fitted plots it doesnt seem to have a perFrctly straight line on 0. I wouldnt say its violated, I would say it is worth looking at more

(B) There is independence between each observation

Based on how the study was done, I believe you can assume Incarity

(C) The errors are normally distributed

Based on Q-Q plot in is mostly normal with some right sqew

(D) The errors have constant variance (homoscedasticity)

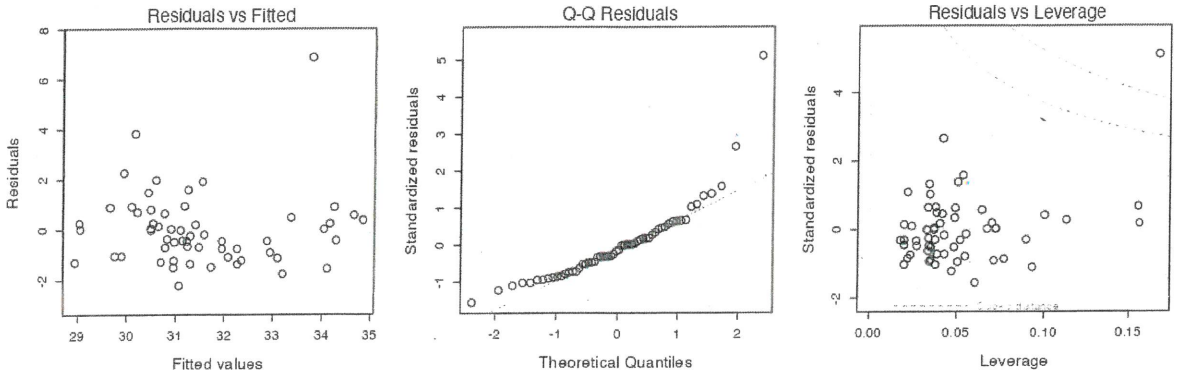
(E) There are no unduly influential points

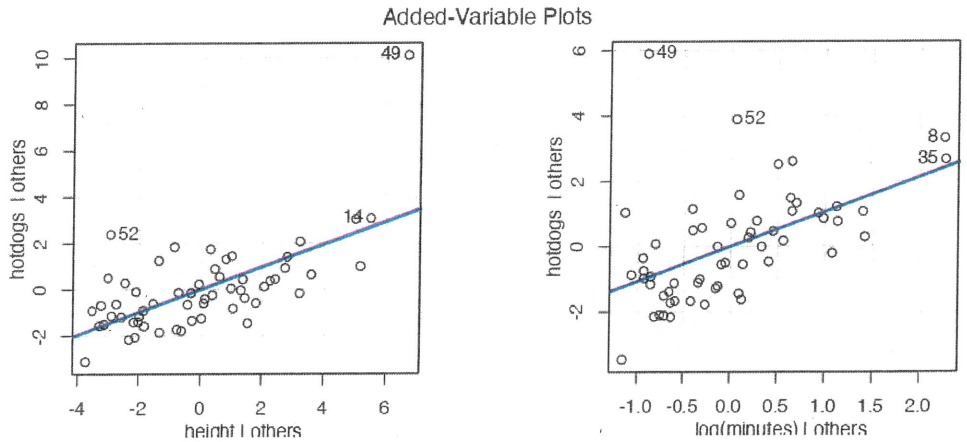
Point 49 could be influence and is worth reviewing

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

M1 M2

BP .071 .028

-I would say while this may be influential in decidind a model, I would not say it is the superior model solely of this test. There are many things to look at suchas CV RSME & Shapiro-Wilk pvalue

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 has a lower pvalue in the Breusch Pagan test but has a higher pvalue inthe Shapiro Wilk Test. It also has a lower CVRSME with all included I think Model 1 is better

Question 16: State the estimated value of β_2 and provide an interpretation in context

$B_2 = 1.06702$

Holding Height Constant a 1 unit increase in minutes is associated with a 106.7% increase in in the conditional mean of Y (hotdogs eaten)

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$68(.4834) + \log(79)(1.0670) + \varepsilon(?)$

$37.53 + \varepsilon$

≈ 37.53 Hotdogs in 10 min

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$37.53 \pm (2.004)(1.485)$

$(34.55406, 40.50594)$

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

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total_length	in cm
tail_length	in cm
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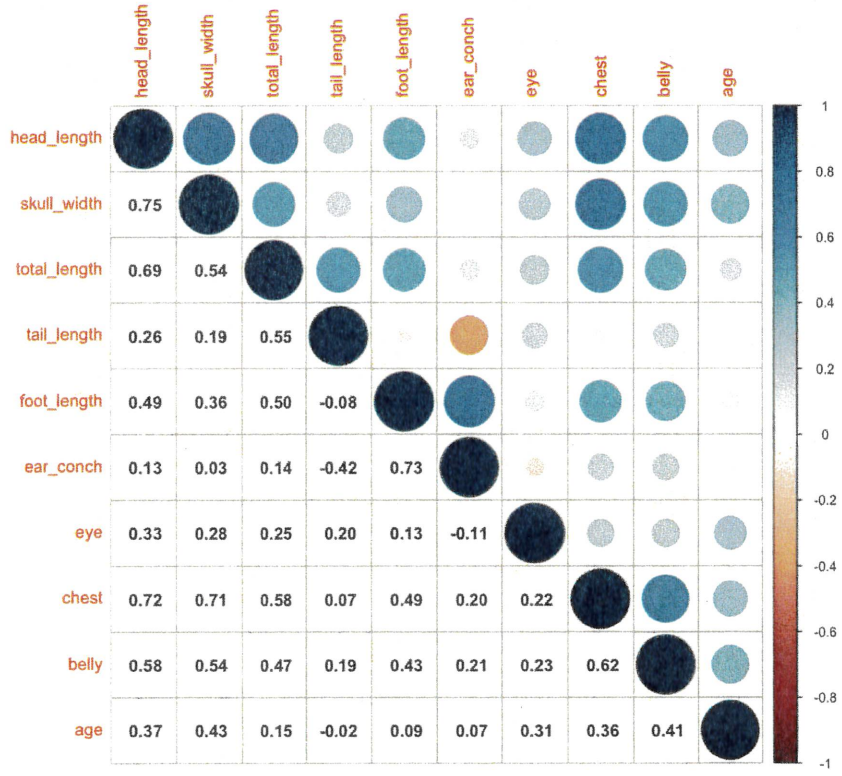
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.16559 \pm 2.01(.10599)$

$.16559 \pm (2.01)(.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true value of the coefficent of belly width is between (-.023, .443)

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

We are seeing that individually

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subset algorithm is a way that we run variables through AIC.

Specifically it does every possible combination and finds the best comba

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

__B__ Average RMSE across all test-training splits and report

__B__ Average the predictions and record their value

__D__ Choose the model with the largest RMSE

__C__ Choose the model with the smallest RMSE

__C__ Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

__A__ Compute each model's AIC and BIC

__A__ Fit each model using the testing data

__A__ Fit each model using the training data

__A__ Initialize vectors for each model that you will use to store their RMSEs

__A__ Refit each model using the full data set and compare the coefficient estimates to the true values

__A__ Split the data into test and training sets

__B__ Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

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Beatty, Kaitlynn

016139318 kbeatty1 808-589-7842

Test: 641422 Form: A
Entry: 07:07pm 10/21/2026
Employee: Diana C.

Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS
40142034
059

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO and ridge regressions are sometimes called shrinkage methods because they help to "shrink" regression model's coefficients. Ridge can shrink coefficients close to zero whereas LASSO can set some coefficients to exactly 0. Shrinkage of coefficients is accomplished through the penalties introduced by LASSO and ridge. LASSO has a penalty $\lambda\sum|B_j|$ whereas ridge has one of $\lambda\sum_j B_j^2$.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is an observable estimate that comes from a sample.
β is the fixed, unobservable population parameter.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

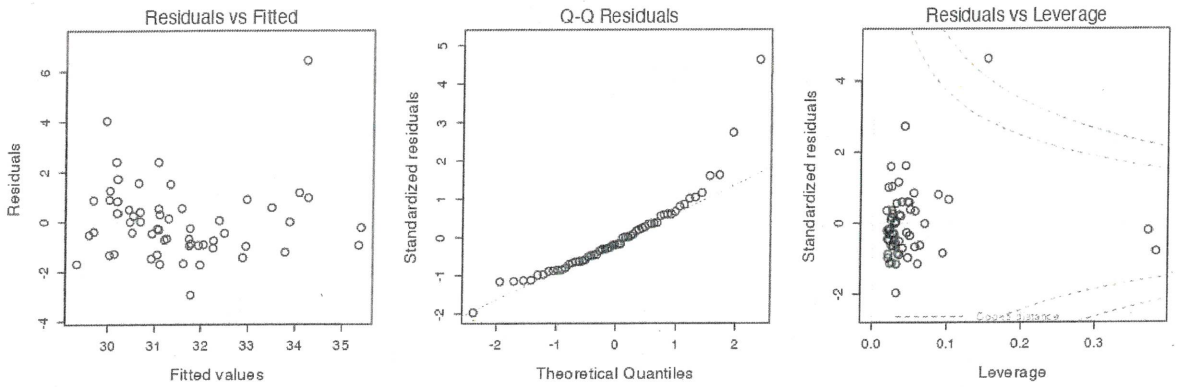
Least squares can be used to estimate regression coefficients because it minimizes each $\hat{\beta}$ to be as small as possible. It helps to find our coefficients and minimize them.

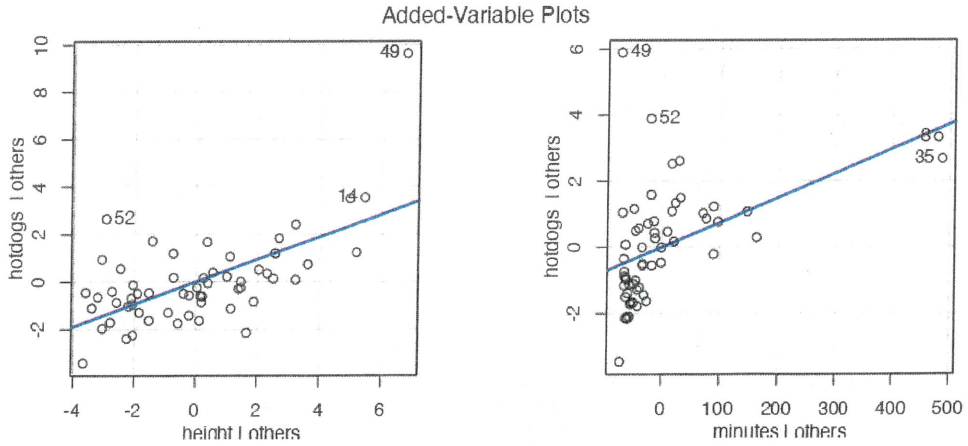
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimated value of B_1 is 0.472892.

Holding minutes constant, a one-inch increase in height is associated with a 0.472892 increase in the average amount of hotdogs eaten.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The height vs. hotdogs plot looks fairly linear while the minutes trained vs. hotdogs have most plots clustered together, with outliers that have surprising values. Because of this, I would say that overall, linearity is not satisfied.

(B) There is independence between each observation

This is satisfied because one contestant's amount of hotdogs eaten doesn't directly influence another contestant's amount of hotdogs eaten.

(C) The errors are normally distributed

Most of the plots in the QQ plot seem to be along the regression line with a few plots being outliers. Overall, normality of errors is satisfied.

Also, the Shapiro-Wilk Test is way below 0.05 (p-value = 2.43e-8).

(D) The errors have constant variance (homoscedasticity)

The Breusch-Pagan p-value is 0.071, which is above 0.05. Therefore, I would conclude that homoscedasticity is not satisfied.

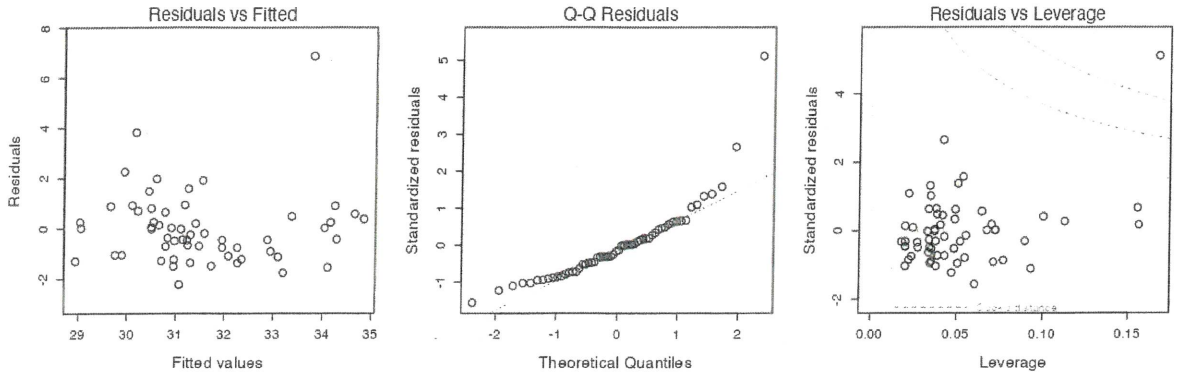
(E) There are no unduly influential points

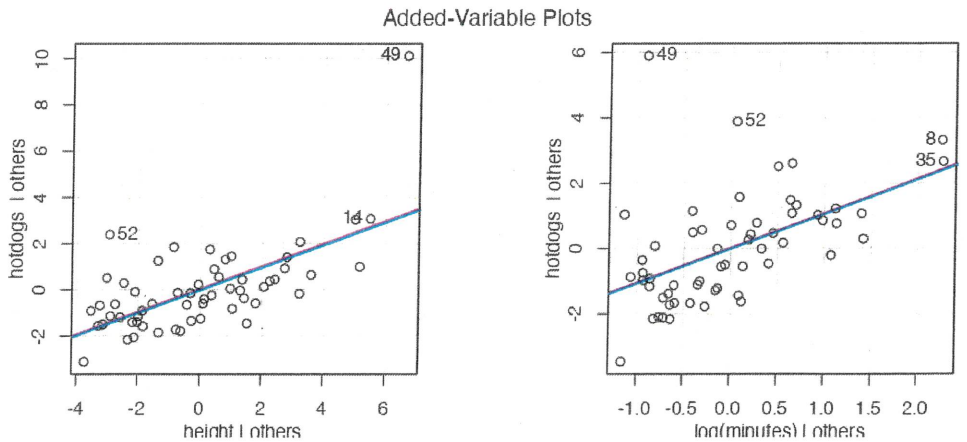
This is satisfied because there are no points that have both high leverage and standardized residual value.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say that looking at the p-values can ignore predictor dependence and joint contributions while producing unstable models. In this case, this only accounts for the equal error variance diagnostic test so we can instead, looking at the other aspects like RMSE, coefficient estimates, t-statistics, etc.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 is a level-log regression model and it is able to make predictions more accurate, adjust the scales, and decrease coefficients that are unnecessarily large. Some disadvantages are that the scale changes for minutes vs. hotdogs so it is no longer in the original response scale. In this case, we'd have to use back transformation to compare predictive accuracy with the model fitted directly to Y.

Question 16: State the estimated value of β_2 and provide an interpretation in context

The estimated value of β_2 is 1.06702.

Holding height constant, a one-minute increase in training is associated with a 1.06702 increase in the average amount of hotdogs eaten.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\text{hotdogs}_i = -6.05589 + 0.48834(68) + 1.06702(4.369)$

$= 31.81$

About 32 hot dogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$32 \pm \left(\frac{0.204}{1.485}\right)(2.004)$

$32 - \left(\frac{0.204}{1.485}\right)(2.004) = 31.725$

$32 + \left(\frac{0.204}{1.485}\right)(2.004) = 32.275$

$(31.725, 32.275)$

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
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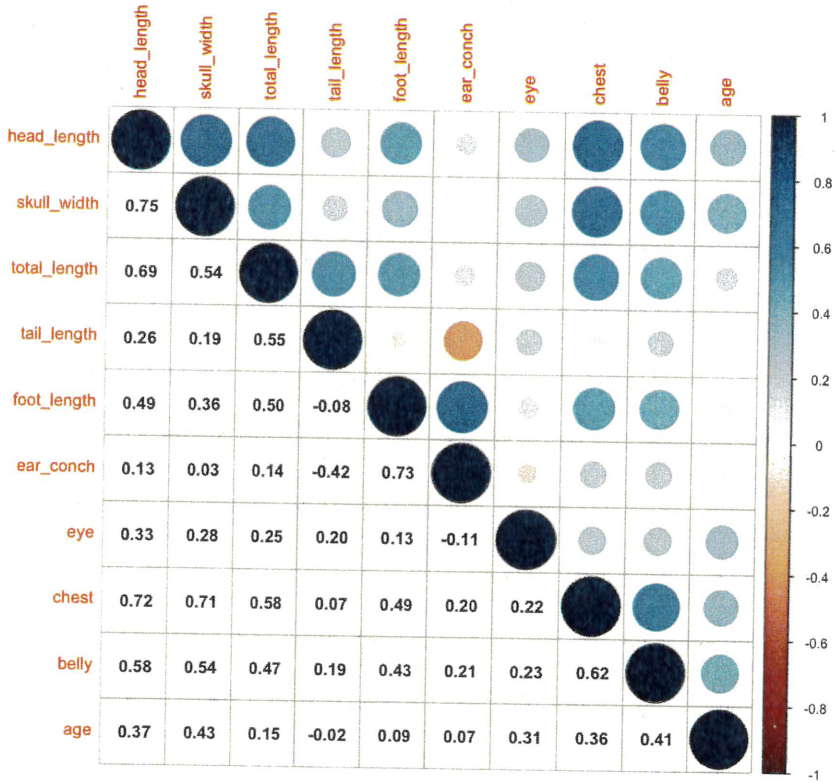
A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

0.16559±2.01(0.10599)

0.16559±2.01(0.10599)

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the average belly width of a local male possum falls between -0.023 centimeters and 0.443 centimeters.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

This "contradiction" is that because there is multicollinearity in the data, it is hard to see how the variables will vary independently. Because of this, the F-test can only detect the variables' joint contributions and not the contribution of each individual variable/coefficient.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets fits every candidate model and keeps the best for the chosen metric. This is a search algorithm, a method or way to choose the best model. After doing so, we can use selection metrics to compare the different models. Using AIC will estimate the expected out of sample information loss as it is a model fit balanced against a complexity penalty.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
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eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

B Choose the model with the largest RMSE

B Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM